

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	176097	49.092977
1	[Git, Change, Create]	51952	14.483372
2	[Git, Change, Delete]	27797	7.749351
3	[Java, ByteCode, Member, Method]	13494	3.761908
4	[Java, ByteCode, Parameter]	13293	3.705872
5	[Git, Commit]	13137	3.662382
6	[File, Git]	11022	3.072754
7	[Git, Change, Rename]	10655	2.970441
8	[Java, ByteCode, Bound, ParameterizedType]	7279	2.029267
9	[Java, ByteCode, Bound]	7240	2.018394
10	[Java, ByteCode, Member, Field]	3609	1.006130
11	[Java, ByteCode, Bound, WildcardType]	2947	0.821576
12	[Java, Value, ByteCode, Annotation]	2931	0.817115
13	[Xml, Element]	2162	0.602730
14	[Java, ByteCode, Member, Method, Constructor]	2120	0.591021
15	[Xml, Text]	1450	0.404236
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.309729
17	[Java, ByteCode, Member, Method, Lambda]	968	0.269863
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.247560
19	[Type, File, Java, ByteCode, Class]	846	0.235851
20	[Json, Key]	702	0.195706
21	[Value, Json, Scalar]	685	0.190967
22	[Java, Value, ByteCode, Primitive]	672	0.187343
23	[Type, File, Java, ByteCode, JavaType]	649	0.180931
24	[Java, ByteCode, Member, Method, GenericDeclar...	578	0.161137
25	[Type, File, Java, ByteCode, ExternalType]	411	0.114580
26	[Git, Change, Copy]	334	0.093114
27	[Author, Git, Person]	299	0.083356
28	[Value, Array]	287	0.080011
29	[Committer, Git, Person]	252	0.070253

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

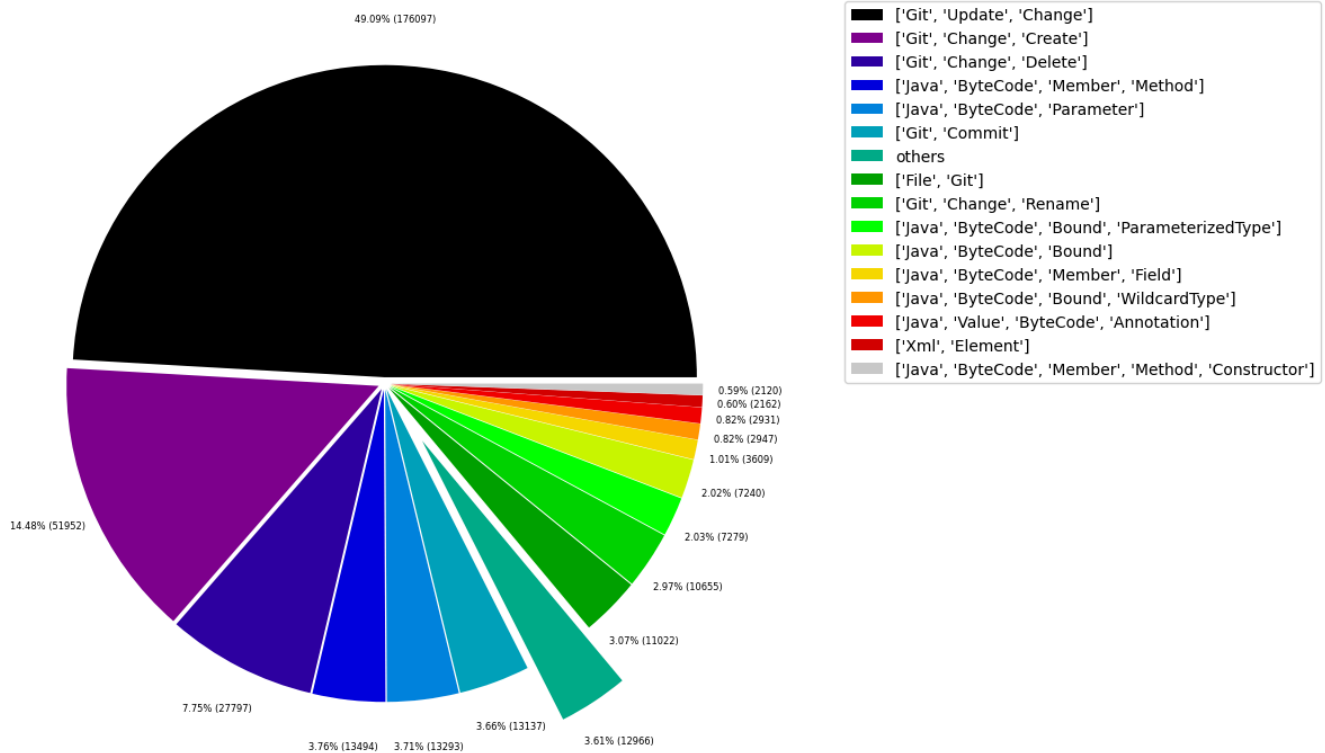


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000279
1	[Package, File, Json, NPM]	1	0.000279
2	[Repository, File, Git]	1	0.000279
3	[File, TS, Scan]	1	0.000279
4	[File, Json]	2	0.000558
5	[File]	3	0.000836
6	[Java, ByteCode, Member, Method, GenericDeclar...	4	0.001115
7	[Maven, Exclusion]	5	0.001394
8	[Value, Array, Json]	6	0.001673
9	[Dependency, NPM]	7	0.001951
10	[Type, File, Java, ByteCode, Void]	9	0.002509
11	[File, Maven, Xml, Pom, Document]	9	0.002509
12	[Java, ManifestSection]	9	0.002509
13	[File, Java, Manifest]	9	0.002509
14	[Artifact, File, Jar, Archive, Zip, Java]	9	0.002509
15	[File, Java, ServiceLoader]	10	0.002788
16	[File, Java, Properties]	12	0.003345
17	[Maven, ExecutionGoal]	16	0.004461
18	[Maven, PluginExecution]	16	0.004461
19	[Xml, Attribute]	18	0.005018
20	[jQAssistant, Rule, Concept]	19	0.005297
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005854
22	[Maven, Configuration]	21	0.005854
23	[Maven, Plugin]	21	0.005854
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006133
25	[Type, File, Java, ByteCode, Enum]	28	0.007806
26	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008364
27	[Xml, Namespace]	36	0.010036
28	[Type, File, Java, ByteCode, Annotation]	44	0.012266
29	[Git, Branch]	46	0.012824

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

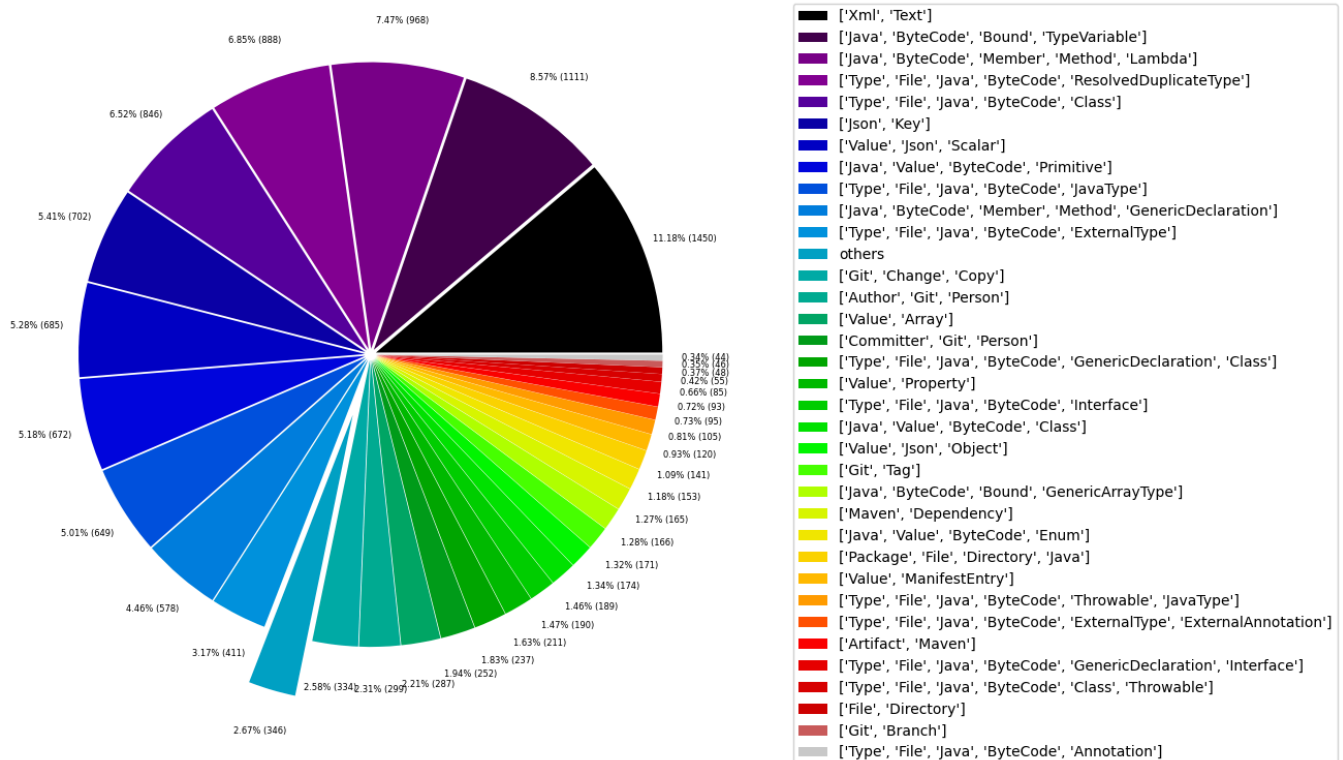


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

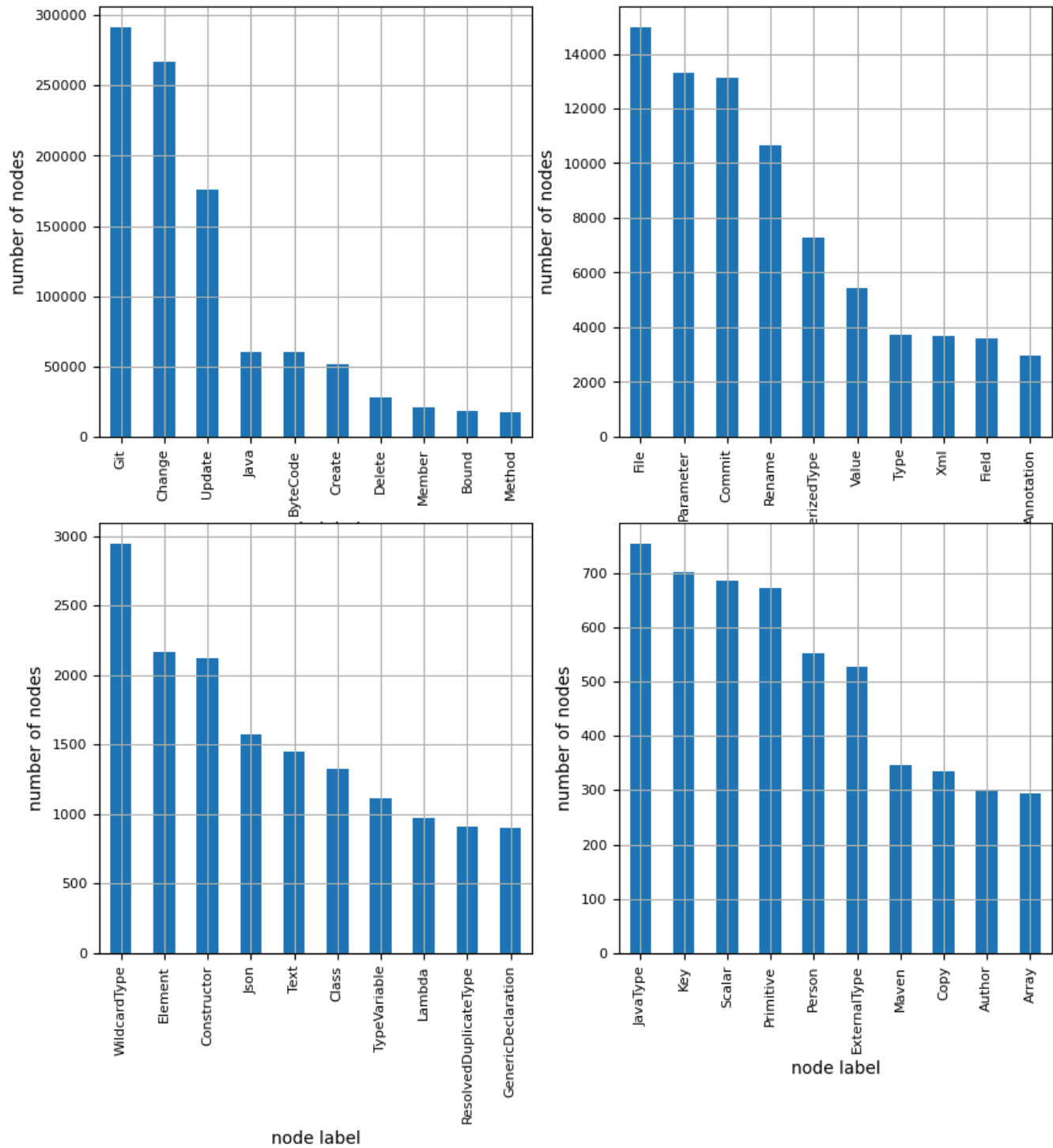
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	291763	81.338775
1	Change	266835	74.389255
2	Update	176097	49.092977
3	Java	60659	16.910742
4	ByteCode	60469	16.857773
5	Create	51952	14.483372
6	Delete	27797	7.749351
7	Member	20773	5.791174
8	Bound	18743	5.225243
9	Method	17164	4.785044
10	File	14983	4.177017
11	Parameter	13293	3.705872
12	Commit	13137	3.662382
13	Rename	10655	2.970441
14	ParameterizedType	7279	2.029267
15	Value	5428	1.513238
16	Type	3715	1.035682
17	Xml	3675	1.024530
18	Field	3609	1.006130
19	Annotation	2975	0.829382
20	WildcardType	2947	0.821576
21	Element	2162	0.602730
22	Constructor	2124	0.592137
23	Json	1570	0.437690
24	Text	1450	0.404236
25	Class	1327	0.369946
26	TypeVariable	1111	0.309729
27	Lambda	968	0.269863
28	ResolvedDuplicateType	910	0.253693
29	GenericDeclaration	904	0.252020
30	JavaType	754	0.210203
31	Key	702	0.195706
32	Scalar	685	0.190967
33	Primitive	672	0.187343
34	Person	551	0.153610
35	ExternalType	527	0.146919
36	Maven	346	0.096459
37	Copy	334	0.093114
38	Author	299	0.083356
39	Array	293	0.081684

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1113059

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	266835	23.973123
1	MODIFIES	266835	23.973123
2	UPDATES	176097	15.820994
3	CREATES	62941	5.654777
4	DELETES	38452	3.454624
5	INVOKES	36776	3.304048
6	COMMITTED	26274	2.360522
7	DEPENDS_ON	22492	2.020737
8	OF_TYPE	21920	1.969348
9	DECLARES	21234	1.907716
10	OF_RAW_TYPE	17353	1.559037
11	HAS_PARENT	15893	1.427867
12	HAS	14429	1.296337
13	HAS_COMMIT	13137	1.180261
14	RETURNS	12844	1.153937
15	HAS_FILE	11022	0.990244
16	RENAMES	10655	0.957272
17	READS	9378	0.842543
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.755306
19	HAS_NEW_NAME	6270	0.563312
20	OF_GENERIC_TYPE	5996	0.538696
21	RESOLVES_TO	5270	0.473470
22	SIMILAR	4078	0.366378
23	WRITES	3944	0.354339
24	CONTAINS	3936	0.353620
25	RETURNS_GENERIC	3593	0.322804
26	ANNOTATED_BY	2919	0.262250
27	REQUIRES	2230	0.200349
28	HAS_FIRST_CHILD	2162	0.194239
29	HAS_LAST_CHILD	2162	0.194239

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

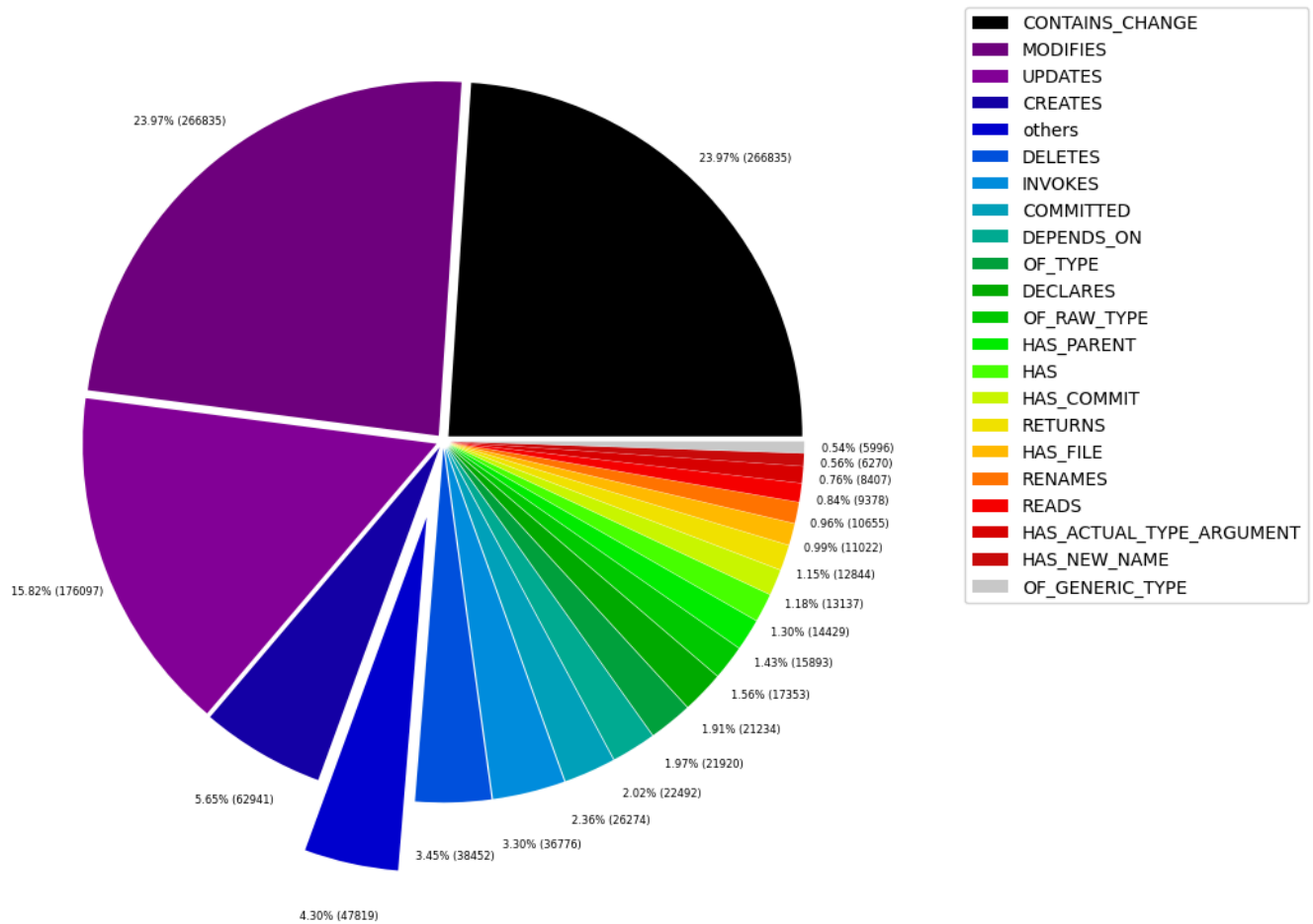


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

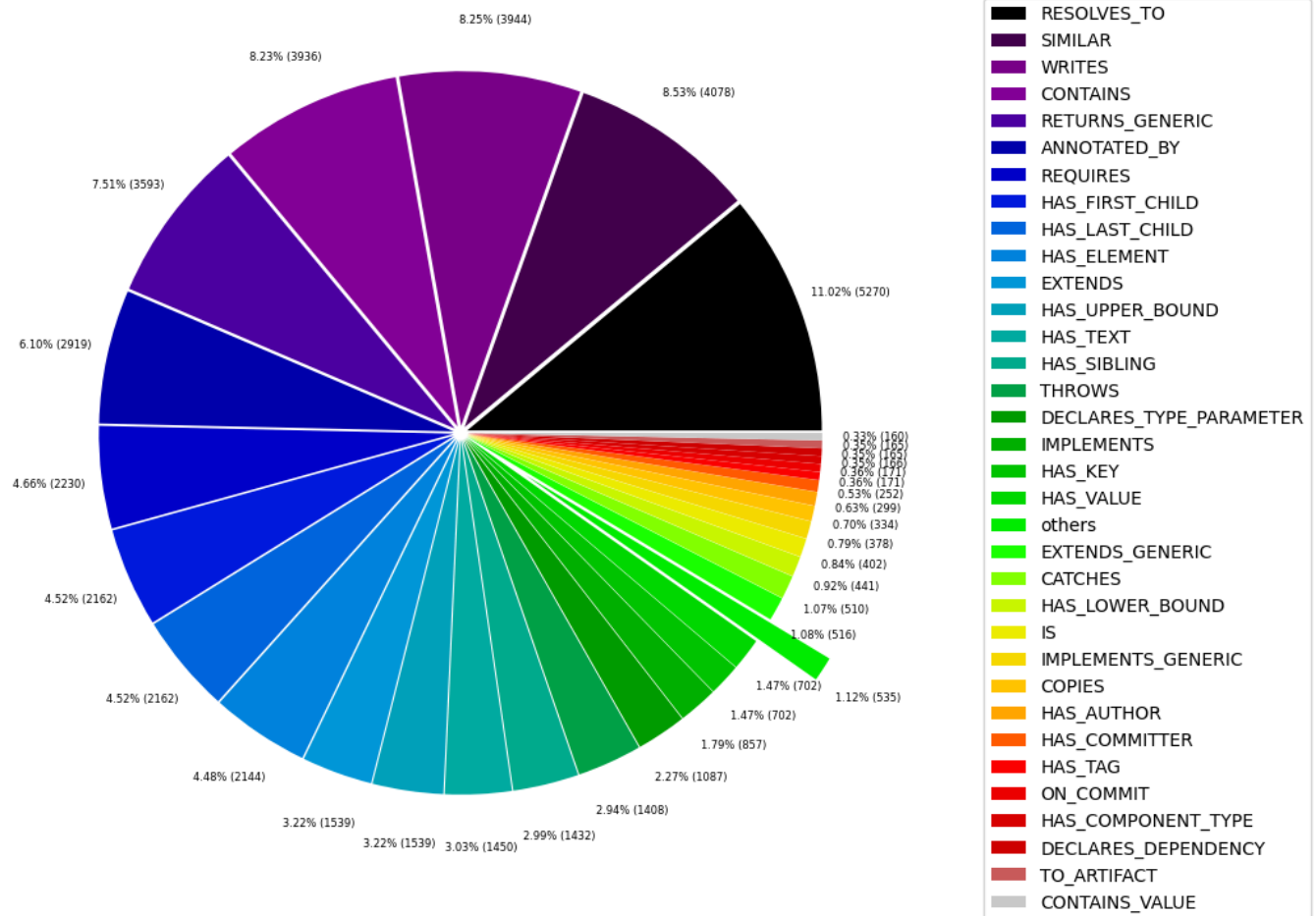
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000090
1	THROWS_GENERIC	5	0.000449
2	EXCLUDES	5	0.000449
3	DECLARES_DEV_DEPENDENCY	7	0.000629
4	DESCRIBES	9	0.000809
5	HAS_ROOT_ELEMENT	11	0.000988
6	HAS_GOAL	16	0.001437
7	HAS_EXECUTION	16	0.001437
8	OF_NAMESPACE	18	0.001617
9	HAS_ATTRIBUTE	18	0.001617
10	INCLUDES_CONCEPT	19	0.001707
11	IS_ARTIFACT	21	0.001887
12	HAS_CONFIGURATION	21	0.001887
13	USES_PLUGIN	21	0.001887
14	REQUIRES_TYPE_PARAMETER	24	0.002156
15	REQUIRES_CONCEPT	28	0.002516
16	DECLARES_NAMESPACE	36	0.003234
17	HAS_DEFAULT	39	0.003504
18	HAS_BRANCH	46	0.004133
19	HAS_HEAD	47	0.004223
20	COPY_OF	127	0.011410
21	CONTAINS_VALUE	160	0.014375
22	DECLARES_DEPENDENCY	165	0.014824
23	TO_ARTIFACT	165	0.014824
24	HAS_COMPONENT_TYPE	166	0.014914
25	ON_COMMIT	171	0.015363
26	HAS_TAG	171	0.015363
27	HAS_COMMITTER	252	0.022640
28	HAS_AUTHOR	299	0.026863
29	COPIES	334	0.030007

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	UPDATES	[File, Git]	176097	176097	
1	[Git, Update, Change]	MODIFIES	[File, Git]	176097	176097	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	176097	13137	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	51952	13137	
4	[Git, Change, Create]	CREATES	[File, Git]	51952	51952	
5	[Git, Change, Create]	MODIFIES	[File, Git]	51952	51952	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	27797	13137	
7	[Git, Change, Delete]	DELETES	[File, Git]	27797	27797	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	27797	27797	
9	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22590	13404	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	15884	13137	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	13137	1	
12	[Committer, Git, Person]	COMMITTED	[Git, Commit]	13137	252	
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	13137	299	
14	[Repository, File, Git]	HAS_FILE	[File, Git]	11022	1	
15	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	10655	13137	
16	[Git, Change, Rename]	CREATES	[File, Git]	10655	10655	
17	[Git, Change, Rename]	RENAMES	[File, Git]	10655	10655	
18	[Git, Change, Rename]	DELETES	[File, Git]	10655	10655	
19	[Git, Change, Rename]	MODIFIES	[File, Git]	10655	10655	
20	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8517	13404	
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8446	13404	
22	[File, Git]	HAS_NEW_NAME	[File, Git]	6270	11022	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156	13293	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7240	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695	13293	
28	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2519	2120	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2474	7279	

Graph Density

```
total_number_of_nodes (vertices): 358701
total_number_of_relationships (edges): 1113059
-> total directed graph density: 8.650759164876725e-06
-> total directed graph density in percent: 0.0008650759164876725
```